

BCA project · cybersecurity

SIEM Security Dashboard

Security Information and Event Management

Cybersecurity Monitoring and Security Event Management System

Introduction & Problem Statement

SIEM collects, monitors and analyzes security events from different sources.



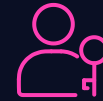
Too many logs

Modern systems generate large numbers of logs. Manual monitoring is slow and difficult.



Repeated failures

Repeated failed logins can indicate brute-force attacks.



Suspicious access

Suspicious file-access activity may indicate unauthorized activity.

A centralized dashboard helps identify and investigate events efficiently.

Objectives & Technologies

Objectives

- Collect and ingest security logs.
- Parse and normalize raw log data.
- Store events in a database.
- Detect suspicious activity using predefined rules.
- Generate severity-based alerts and track source IPs.
- Provide search and filtering.
- Display information in an interactive dashboard.

Technologies



Backend — Python, FastAPI



Storage — SQLite, SQLAlchemy



Frontend — React, Vite, Axios

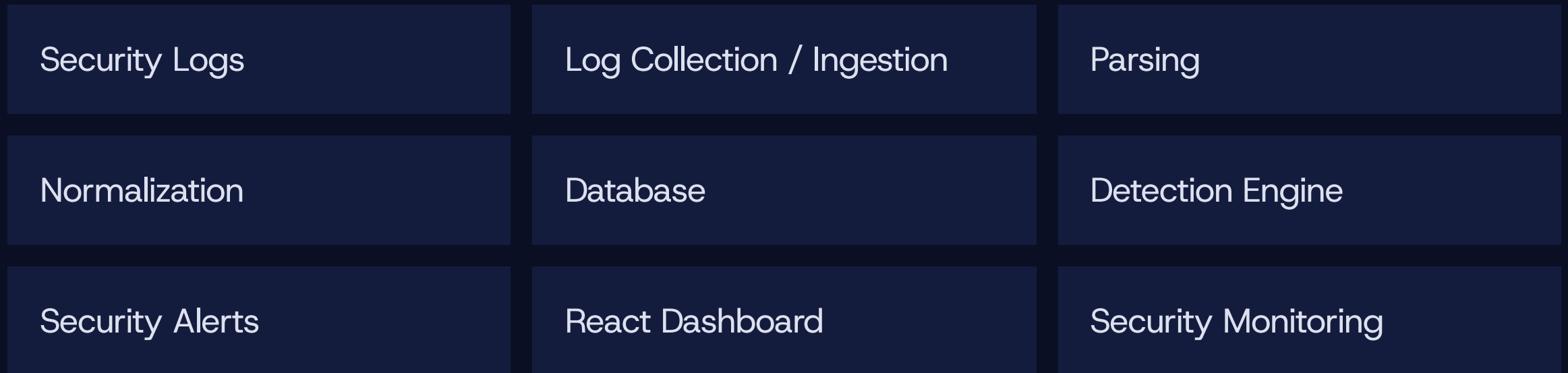


API documentation — Swagger UI

System flow · from raw logs to monitoring

SIEM Architecture & Working

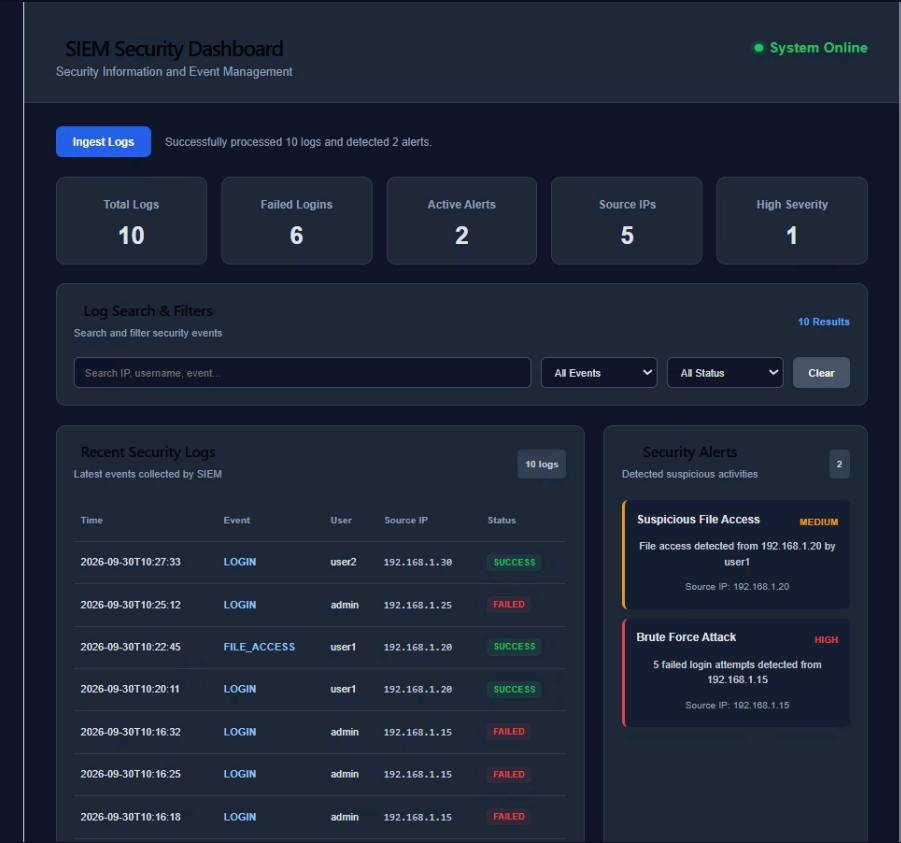
Logs are collected, converted into structured data, normalized, stored, analyzed by detection rules, then shown as alerts on the dashboard.



Brute-force detection rule

5 or more failed login attempts from the same IP → HIGH severity Brute Force alert.

Main SIEM Dashboard



Total Logs: 10

Failed Logins: 6

Active Alerts: 2

Source IPs: 5

High Severity: 1

A centralized view of security events, failed logins, source IPs and generated alerts.

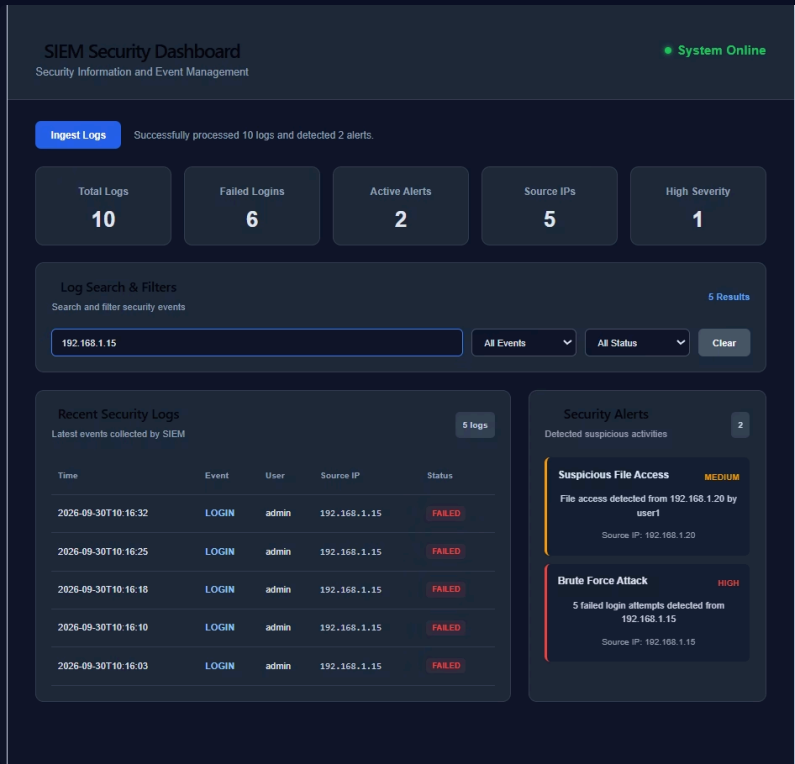
Log Ingestion & Security Alerts



- Sample logs are parsed into structured fields and normalized.
- Detection rules analyze processed events.
- 10 logs processed successfully; 2 alerts generated.
- Brute Force Attack, HIGH: 192.168.1.15, 5 failed login attempts.
- Separate Suspicious File Access alert, MEDIUM severity.

Investigation · narrow the event set

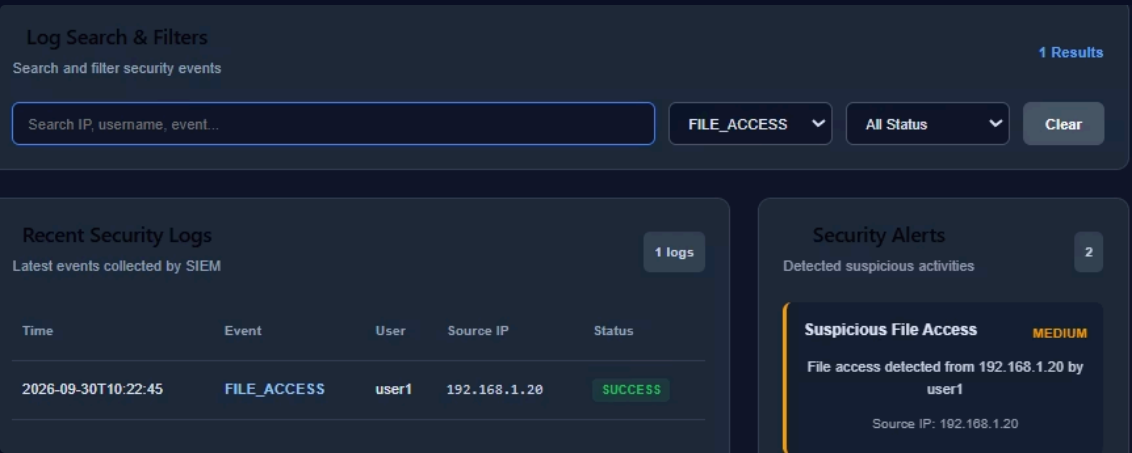
Search & Event Filtering



Search 192.168.1.15

Shows the 5 related failed-login events for investigating one source IP.

Filtering helps analysts narrow down relevant events quickly.



FILE_ACCESS filter

Isolates one event: user1, 192.168.1.20, SUCCESS.

Backend API Documentation

SIEM Security Dashboard 1.0.0 OAS 3.1

/openapi.json

Cybersecurity monitoring and security event management system

Logs ^

POST	/logs/ingest	Ingest Logs	▼
GET	/logs/	Get Logs	▼
GET	/logs/alerts	Get Alerts	▼

default ^

GET	/	Root	▼
GET	/health	Health Check	▼

- POST /logs/ingest · Ingest Logs
- GET /logs/ · Retrieve Logs
- GET /logs/alerts · Retrieve Alerts
- GET / · Root API endpoint
- GET /health · Health Check

FastAPI Swagger UI provides interactive API documentation and allows backend endpoints to be tested directly.

Key Features & Future Scope



Key features

- Log ingestion, parsing and normalization
- Database storage
- Rule-based brute-force and suspicious file-access detection
- Severity classification and source IP tracking
- Search and event filtering
- Alert generation, interactive dashboard, REST API



Future scope

Not yet implemented

- Real-time log ingestion
- Additional detection rules
- Threat intelligence integration
- Authentication and role-based access control
- Advanced analytics and charts
- Email/webhook notifications
- Advanced anomaly detection

Project outcome

Conclusion

The basic SIEM workflow is successfully demonstrated.



Process

Collects, processes and normalizes security logs.



Detect

Predefined rules identify suspicious activity and generate severity-based alerts.



Investigate

Source-IP tracking, search and filtering are presented in a centralized dashboard.

Future extensions can support real-time SOC monitoring and advanced threat detection.